EduVision: A Smart AI-powered Web App for Visually Impaired Learners

📘 1. User Guide

🔹 How to Use:

1. **Upload Educational Material:**
   - Click on the "Upload" button to upload any educational file (.pdf, .jpg, .png, .ppt, .pptx, .docx, or .txt).
2. **Choose the Input Type:**
   - Text Extraction: Extract and display readable text from uploaded material.
   - Audio Output: Convert the extracted text into natural-sounding speech using gTTS.
   - Summarization: Generate a concise summary with keyword highlights for improved understanding.
3. **Interact with the Output:**
   - **Extracted text is shown in an output section.**
   - **Click "Listen" to hear the text read aloud.**
   - **View the summary and key terms generated for quicker comprehension.**

💡 ***Tip: EduVision runs in your browser — no installation required. Works on both desktop and mobile devices.***

🛠️ 2. Technical Documentation

🔹 Overview

EduVision is built to enhance educational access for visually impaired learners by transforming diverse documents into accessible formats: Text, Audio, and Summarized Content.

🔹 System Architecture

- Frontend & Backend: Built entirely using Streamlit (single streamlit_app.py).
- OCR (Optical Character Recognition): Uses Tesseract OCR to extract text from images or scanned documents.
- Text-to-Speech (TTS): gTTS (Google Text-to-Speech) converts extracted text into audio.
- Summarization & Keyword Extraction: Implemented using SpaCy and TextRank algorithms.

🔹 Technologies Used

Tool/Library	Purpose
Streamlit	Web UI and backend logic
Tesseract OCR	Extract text from images and PDFs
gTTS	Convert text into human-like speech
SpaCy + TextRank	Summarize text and extract keywords
Python	Core application logic
Streamlit Cloud	Deployment platform
GitHub	Version control

🔹 **Process Flow**

1. User uploads a document.
2. EduVision extracts readable text using Tesseract (if applicable).
3. The text is summarized and key phrases are extracted.
4. The user can view the text, summary, and listen to the audio output.

🧾 3. Code Walkthrough

🔹 **File Structure**

📁 **eduvision-streamlit/**

└── 📄 **streamlit_app.py**

🔹 **streamlit_app.py Responsibilities**

- Handles file uploads, OCR, text extraction, summarization, text-to-speech, and all UI rendering.
- Implements logic for different file formats using conditional checks.
- Uses Python libraries like pdfplumber, pytesseract, docx2txt, python-pptx, spaCy, and gTTS.

👤 4. User Stories

📌 **1. *Visually Impaired Student – Maria***

Maria uploads her study material to EduVision and listens to her book being read aloud. She uses the summarization to focus on key sections without needing assistance.

📌 2. ***Teacher in a Rural Area – Mr. Sam***

Mr. Sam uses EduVision to convert his handwritten lesson plans into audio and summary formats, making education accessible for his visually impaired students.

⚠️ 5. Limitations & Future Work

🔹 Limitations

- Language Support: Currently optimized for English.
- OCR Sensitivity: Accuracy depends on image quality.
- Performance Lag: Processing time increases for large files.

🔹 Future Work

- Add multi-language support for text and speech.
- Integrate voice commands for hands-free navigation.
- Enhance summarization for complex documents.
- Improve accessibility and support for more assistive tools.

🔗 6. GitHub Repository (README Summary)

EduVision – An AI-powered platform that:

- Converts documents and images into text and audio.
- Summarizes content with keyword extraction.
- Requires no installation, accessible through any browser.

GitHub Repo: https://github.com/Arthi-2004/eduvision-streamlit

🚀 **Installation Steps:**

git clone https://github.com/Arthi-2004/eduvision-streamlit.git

cd eduvision-streamlit

pip install -r requirements.txt

streamlit run streamlit_app.py